Name: _____ Class and student number: _____

Check answers and correct

Give out C only after B is completed. Check, keep the first answers and correct them. Do not invent an original hypothesis.

Item	Check points; other sound wording is accepted
B1.1	Oil is above the water, with a clear boundary. Keep the visible description separate from a chemical explanation.
B1.2	The prediction can be tested, but the reason is wrong. Y is lipase (脂肪酶), which speeds up hydrolysis. Bile (膽汁) causes emulsification (乳化作用). Testable does not mean already proved.
B1.3	Independent: combination of juices. Dependent: appearance. Controlled: starting oil/water volume, temperature, time, total added volume and shaking method.
B1.4	Choose equal time, equal starting and total added volumes, and equal shaking; do not choose extra total liquid for X + Y.
B2.1	The 2 mL water set is the baseline for this question. Equal added volume avoids a difference in total volume or dilution.
B2.2	In order: water control, 2 mL water; X, 2 mL; Y, 2 mL; X + Y, 1 mL each.
B2.3	Unfair: 2 mL versus 4 mL. Use 1 mL of each juice, keeping the total at 2 mL to avoid a volume/dilution difference.
B3.1	Water: separated layers; X: cloudy; Y: thinner oil layer, still separated; X + Y: clear.
B3.2	Cloudy emulsion (乳狀液). X disperses oil droplets by emulsification (乳化作用), a form of physical digestion (物理消化), without changing the lipid molecules.
B3.3	Increases. For the same total amount of lipid, small droplets provide more surface for lipase (脂肪酶) to act on. This is not a measurement of speed.
B4.1	Choose the statement that appearance gives clues but does not prove glycerol (甘油) and fatty acids (脂肪酸) formed. Other chemical product tests and repeated tests are needed.
B4.2	The unboiled X + Y set, 1 mL each; cool the boiled juice to the common 37°C and keep the other conditions equal. This control contains X and Y.
B4.3	Boiled X + unboiled Y: clear; unboiled X + boiled Y: cloudy.
B4.4	Y. Lipase (脂肪酶) is a protein; boiling causes denaturation and loss of function. Cooling does not restore it in this model.
B4.5	For example, compare water with X for emulsification (乳化作用), and unboiled X + Y with boiled Y for enzyme loss. Explain bile (膽汁), lipase (脂肪酶) hydrolysis and heat denaturation. Appearance and different individual juice amounts limit measured claims. Other sound specific comparisons are accepted.

Correction 1

Choose an answer in B to correct. Write its ID, the sound explanation and why it was wrong. If all are correct, add evidence and a limit to one answer; do not invent a mistake.

Correction 2

Use one named comparison to explain emulsification (乳化作用), enzyme action or denaturation in your own words.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.